IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- The project aims to analyze SpaceX's Falcon 9 rocket launch data and to develop machine learning models to predict whether the first stage will be safely landed after flight.
- Summary of methodologies
  - The data were extracted from SpaceX and various online sources and were then processed for model development. Data exploration were done through visualizations like static and interactive charts, geographical mapping and database queries.
- Summary of all results
  - Landing the first stage of the rockets became more successful as time went on due to ongoing research and technological advancements. All the classification models tested scored 83.33% accuracy regardless of algorithm which may be due to the limited test data of only 18 samples.

Introduction

- Project background and context
  - SpaceX's Falcon 9 launch rocket revolutionized the rocket industry by introducing the reusability of its stage 1—safely landing back to earth.
    - Unlike the single-use stage 1 rockets of the previous era
    - More cost-effective and cost-efficient
  - Being at the forefront, SpaceX is a crucial source of data to learn and gain insights from in order to be competitive in the market.
- Problems to answer
  - Can a model predict the success of Falcon 9's first stage landing?
  - Which factors possibly influence or can be good predictors of the success rate?
- SpaceY vision: To develop our own cheaper and efficient space launch missions

Section 1

Methodology

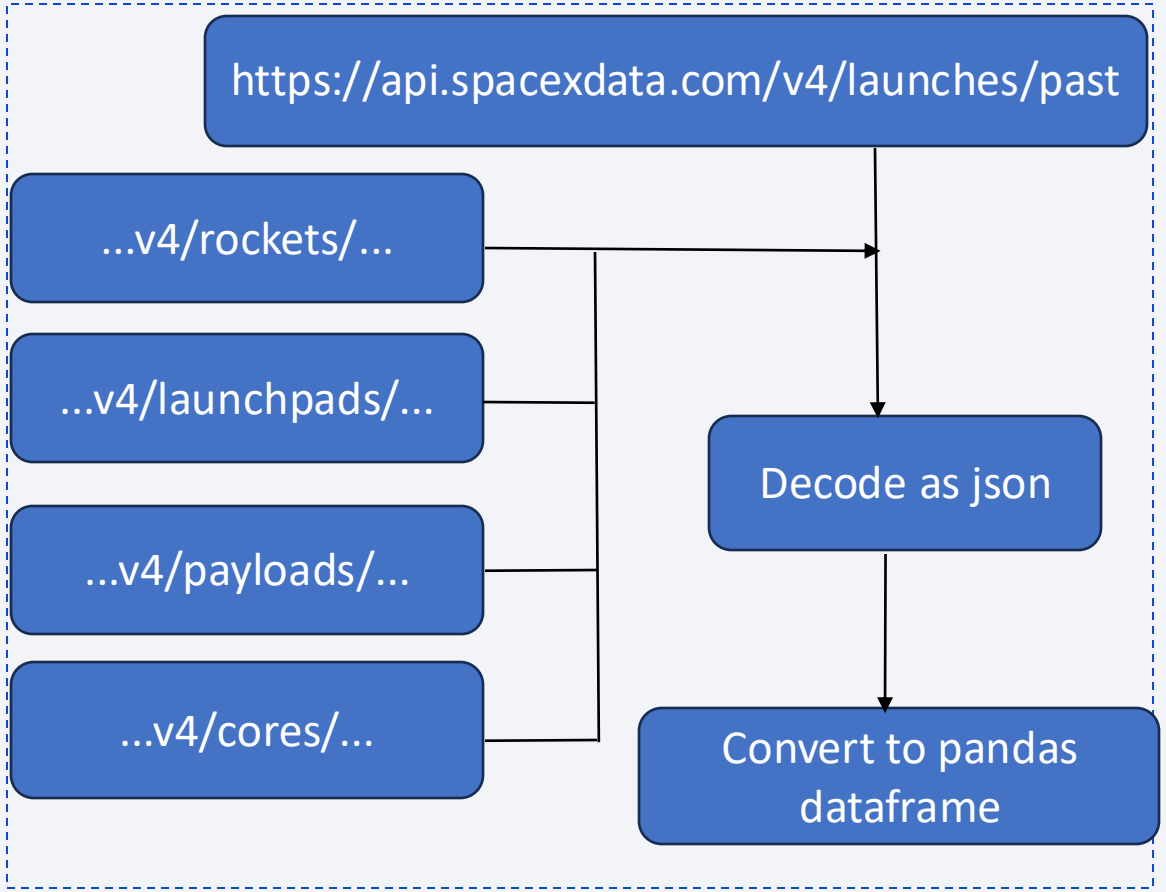
Methodology

Executive Summary

- Data were collected utilizing the SpaceX API and web scraping SpaceX's historical launch records.
- Data were cleaned, missing values were imputed and categorical variables one-hot encoded.
- Exploratory data analysis were performed using SQL queries and visualizations such as plots, Folium maps and interactive Plotly dashboards.
- Predictive modelling of outcome (success or failure of stage 1 landing) was performed using classification models (e.g. logistic regression, decision tree classification), tuned via GridSearch CV, and tested using the unseen 20% of data.

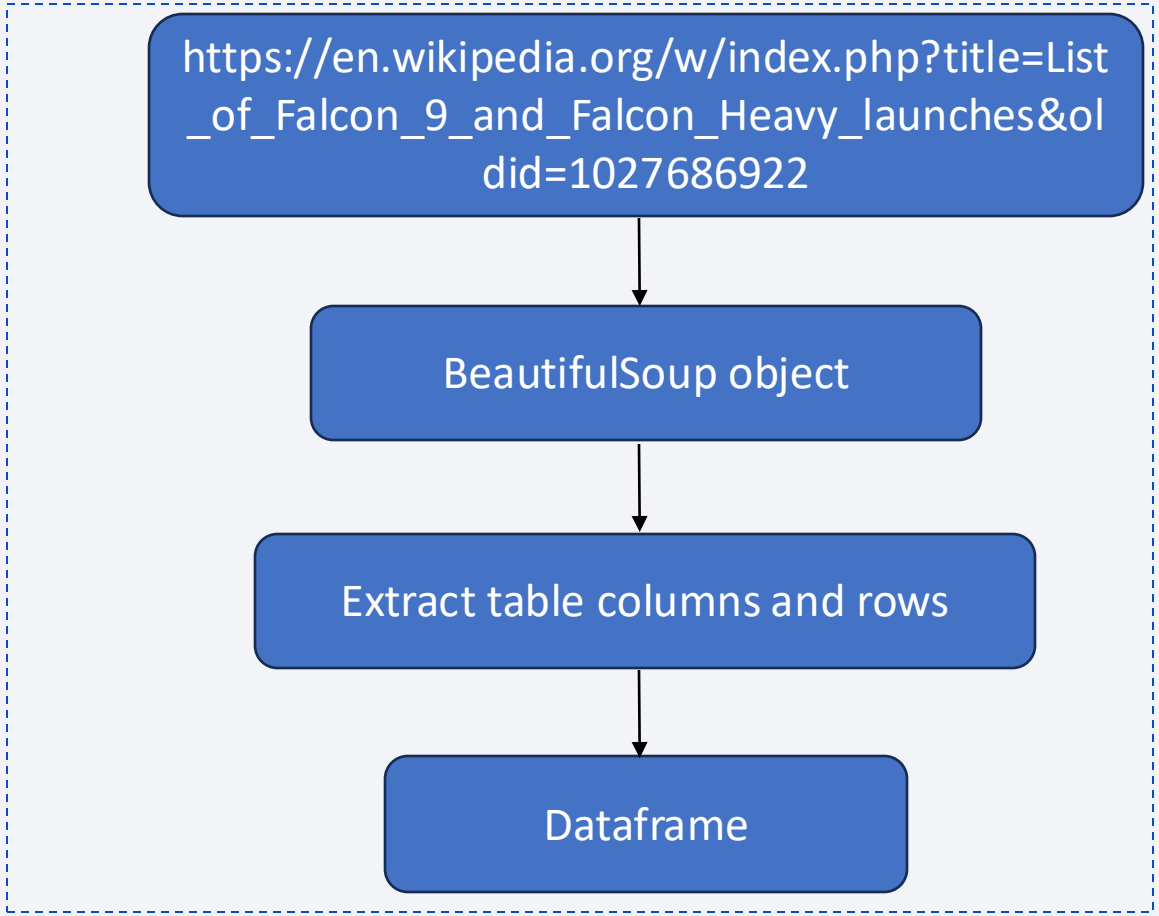
Data Collection – SpaceX API

- GET requests to the SpaceX API through the following URLs (referring to past launches, rockets, launchpads, payloads, cores)
- Data combined into a dataframe
- Filtered for Falcon 9 launches only



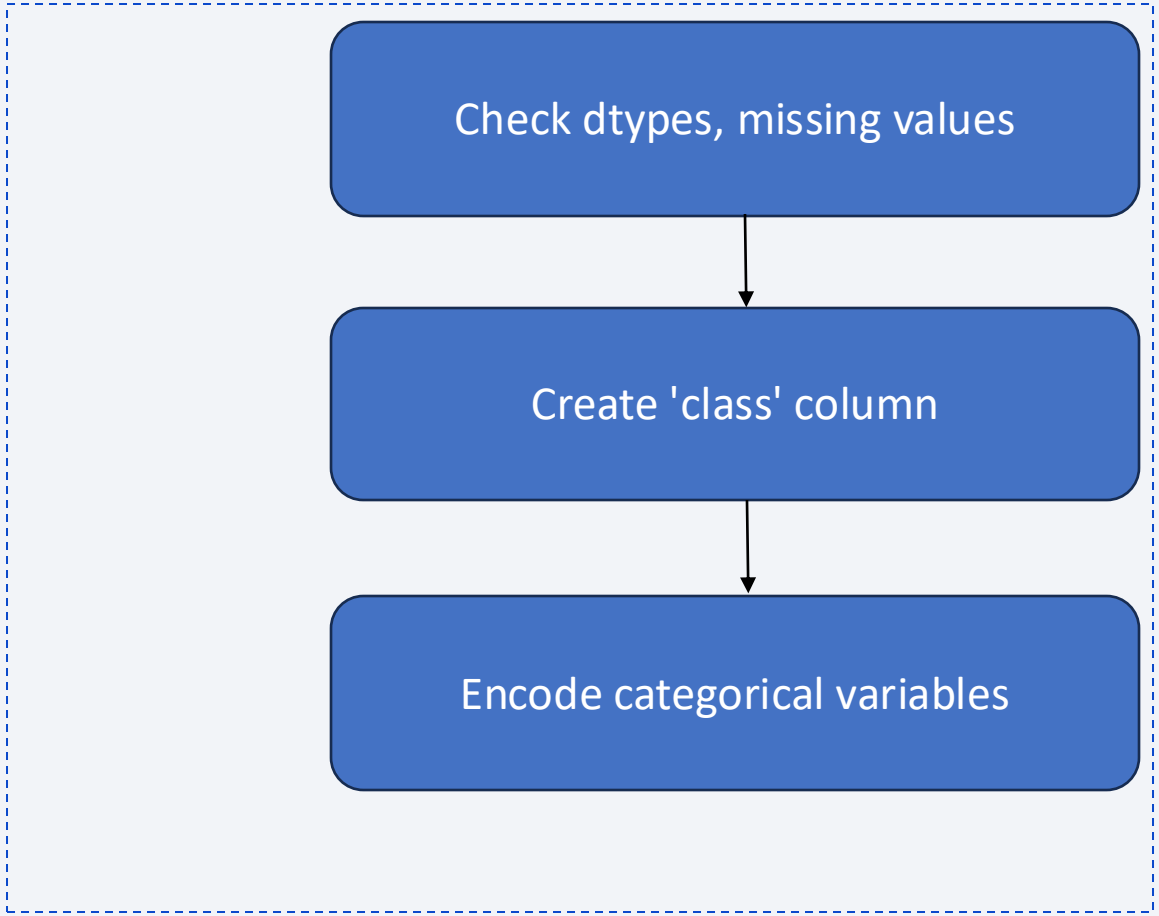
Data Collection - Scraping

- GET request to the Falcon 9 Launch Wiki page (2021)
- BeautifulSoup was used to extract data from the html response



Data Wrangling

- 'Outcome' column was transformed into 'class' column—the training label or the target variable
  - 0 means failed stage 1 rocket landing
  - 1 means successful stage 1 rocket landing
- Categorical variables were one-hot encoded
- Numeric variables



EDA with Data Visualization

The following plots were created to visually assess the relationship between them and with the target variable 'class'

- Flight number vs payload mass scatter plot
- Flight number vs launch site scatter plot
- Payload mass vs launch site scatter plot
- Landing success rate vs orbit type bar chart
- Flight number vs orbit type scatter plot
- Payload mass vs orbit type scatter plot
- Yearly success rate line plot

EDA with SQL

- Exploratory queries were performed such as getting the distinct launch site names, the sum of payload mass launched through NASA (CRS) missions, booster version names with particular achievements, counts of success and failed missons

Build an Interactive Map with Folium

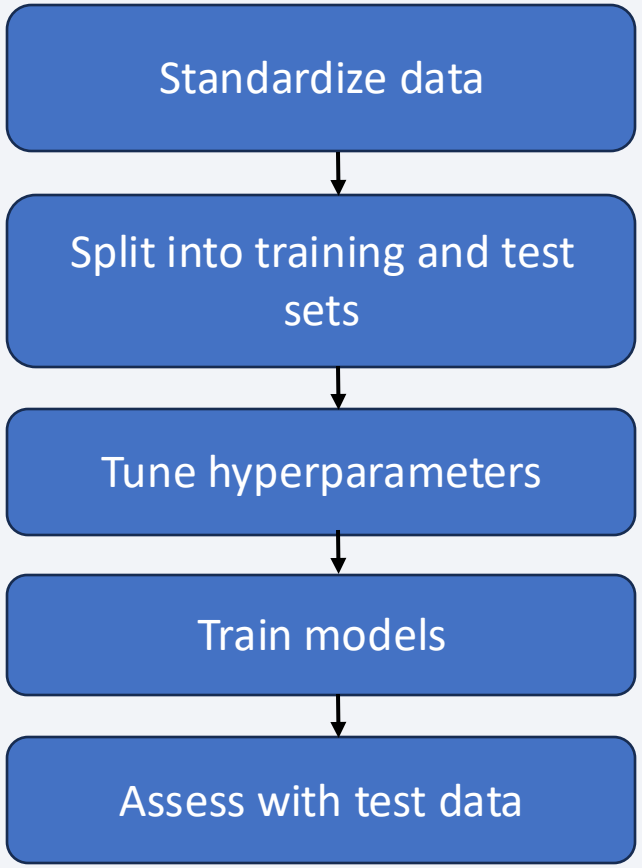
- An interactive map was built with Folium to analyze geospatial data
- Launch sites were circled and labeled. Into each, launch outcome markers were also added.
- Distance lines and measurements between a selected site and its nearest coastline, highway, railroad and city were added to assess the proximity of the site.

Build a Dashboard with Plotly Dash

- Launch site drop-down input component to filter the dashboard visuals
- Pie chart to show the breakdown of the total successes by launch site when 'all sites' were selected, and the proportion of outcomes from each selected site
- Payload mass range slider to assess visuals within a selected range
- Scatter plot to show the outcome (class) vs the payload mass with range based on the slider

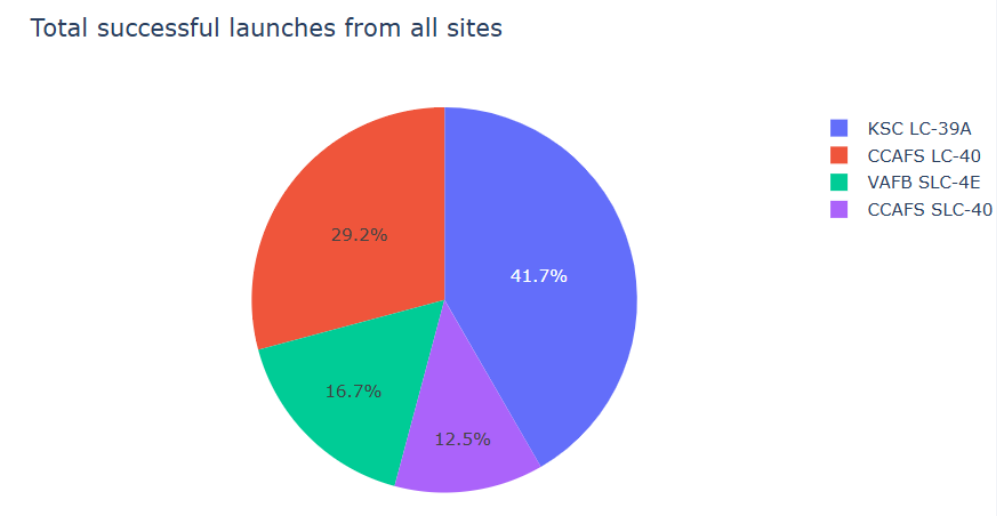
Predictive Analysis (Classification)

- Data was standardized using StandardScaler()
- Split into training and test (20%) sets using train_test_split()
- Model hyperparameters were tuned with GridSearchCV then fitted to the training data
- Model performance was then assessed with the test data
- Confusion matrix was also examined



Results

- From EDA – booster landing improved as time went on; there is no one feature clearly correlated with landing success
- From interactive analytics



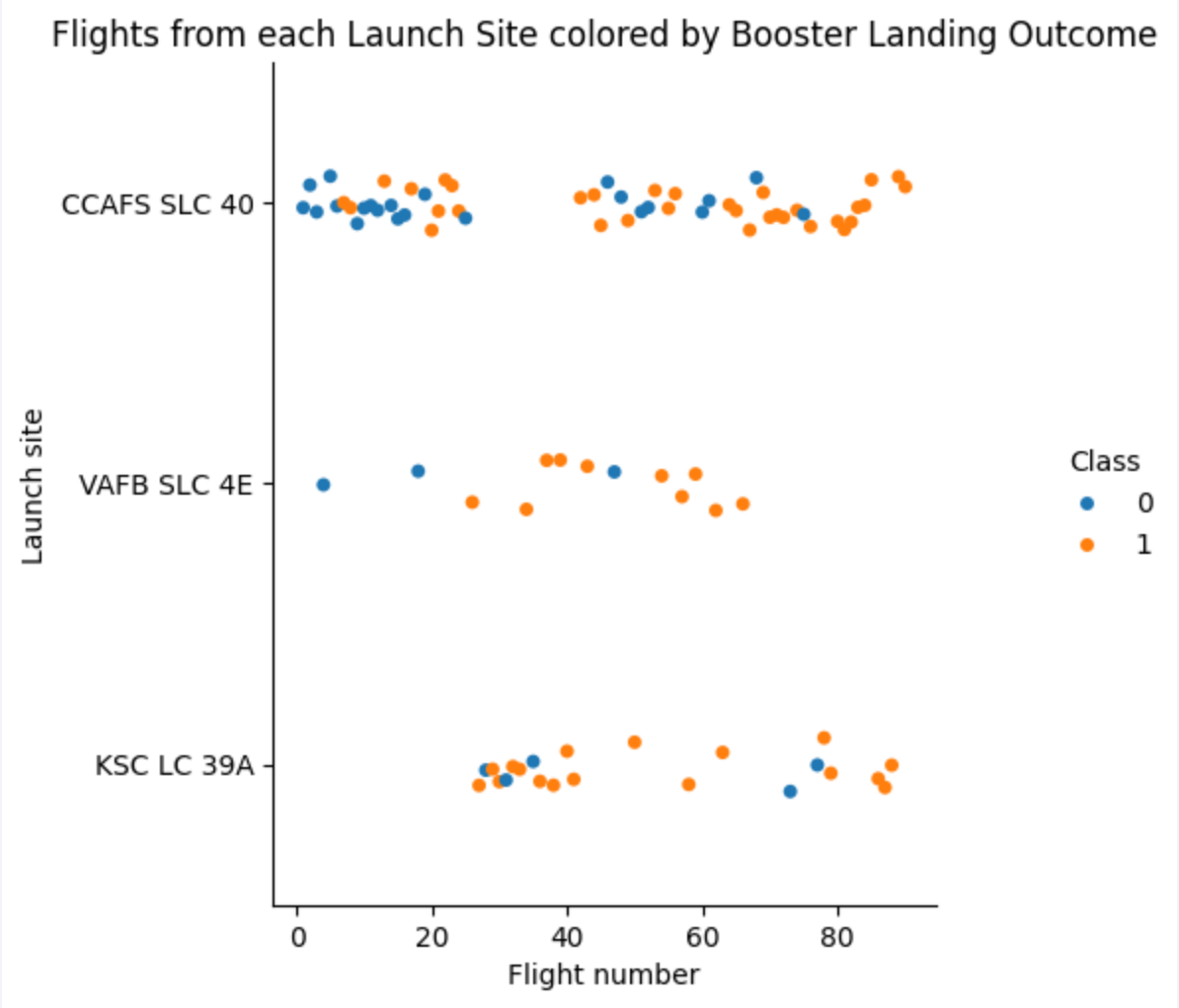
- From predictive analytics – all tested algorithms (logistic regression, SVM, KNN, Decision tree) had test accuracy of 83.33%

Section 2

Insights drawn from EDA

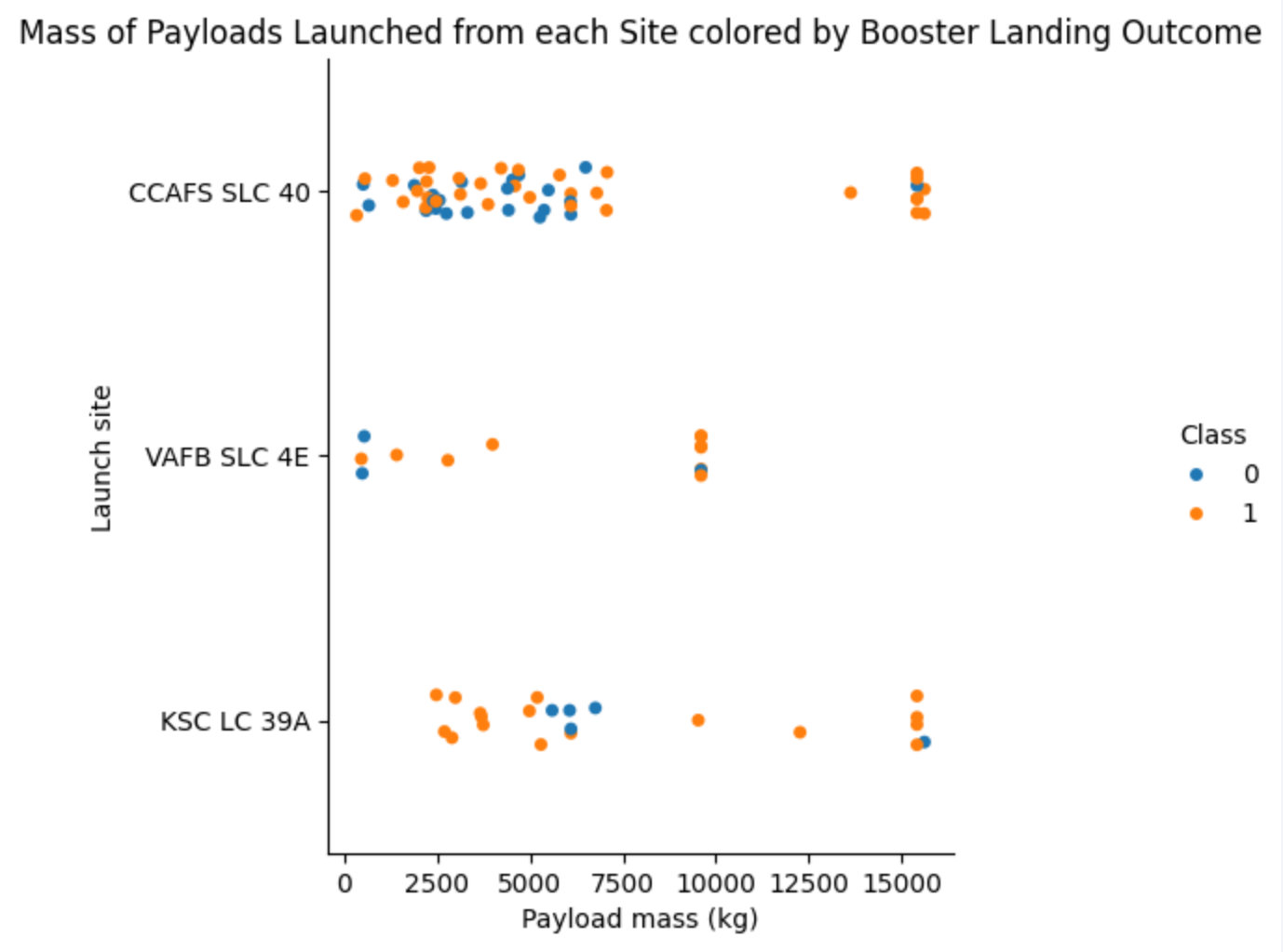
Flight Number vs. Launch Site

- Later flights were more successful in landing the Falcon 9 boosters than the earlier ones.
- Majority of flights were launched from Cape Canaveral (CCAFS)



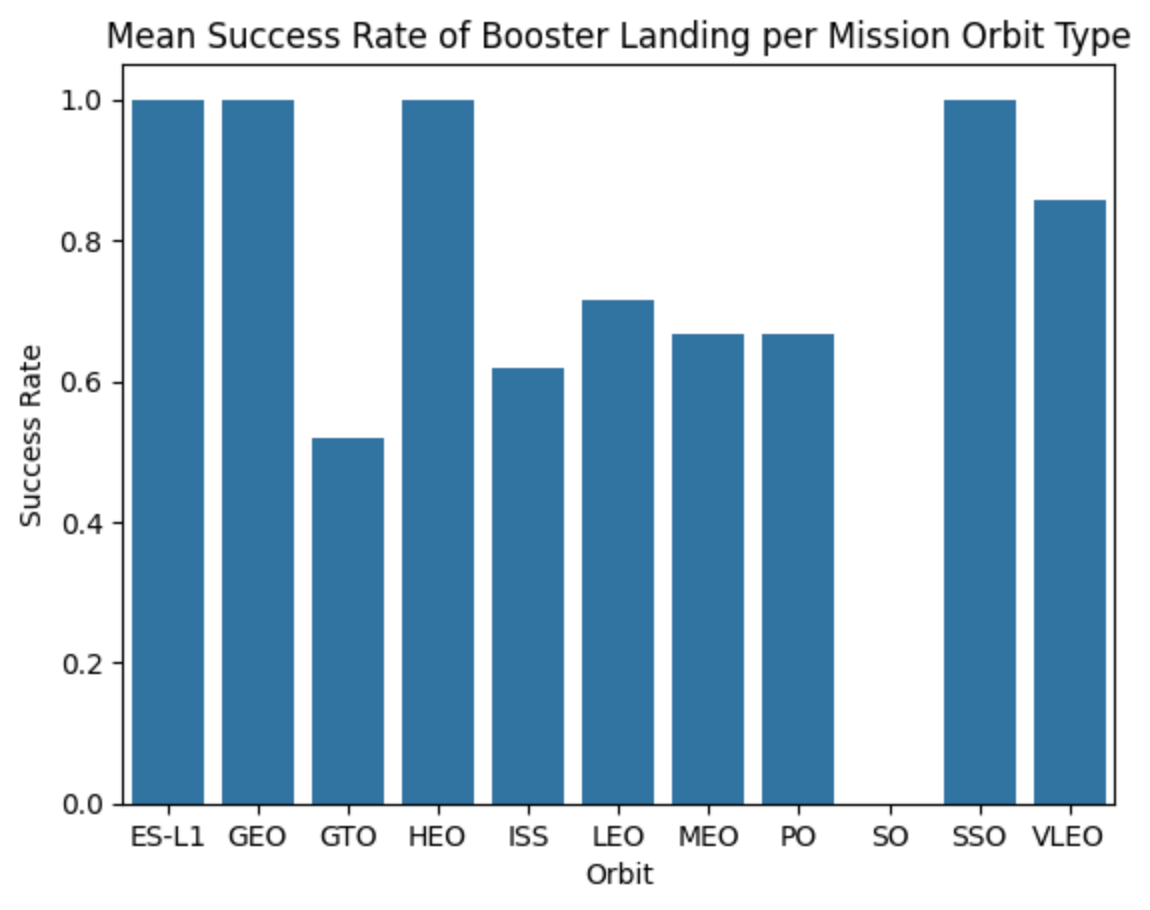
Payload vs. Launch Site

- The success rate of landing the Falcon 9 boosters was fairly equal across the range of payload masses, regardless of launch site.
- The very heavy payloads were launched from from CCAFS and KSC sites.



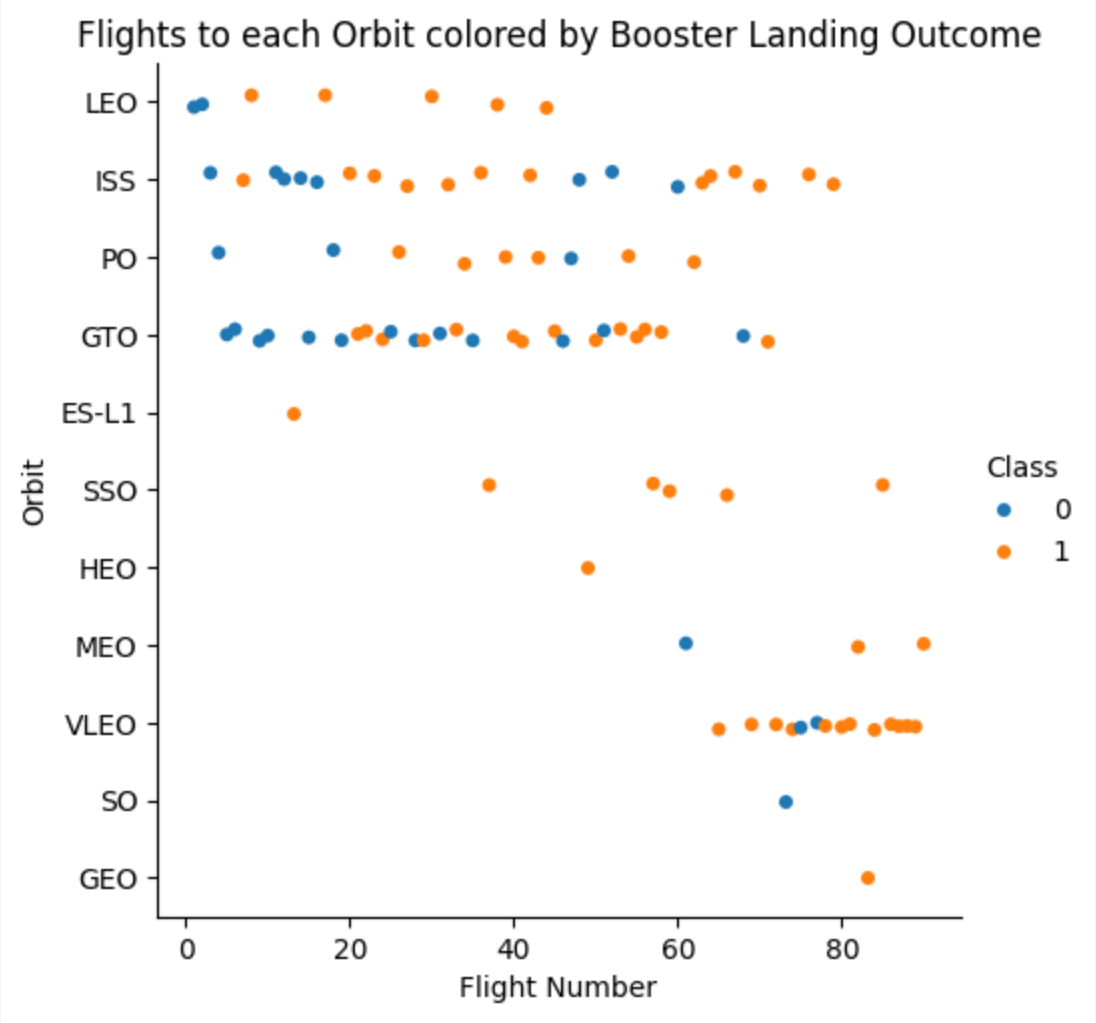
Success Rate vs. Orbit Type

- The success rates of flights to ES-L1, GEO, HEO, SSO orbits were perfect.
- The lowest is to GTO orbit at 52%.



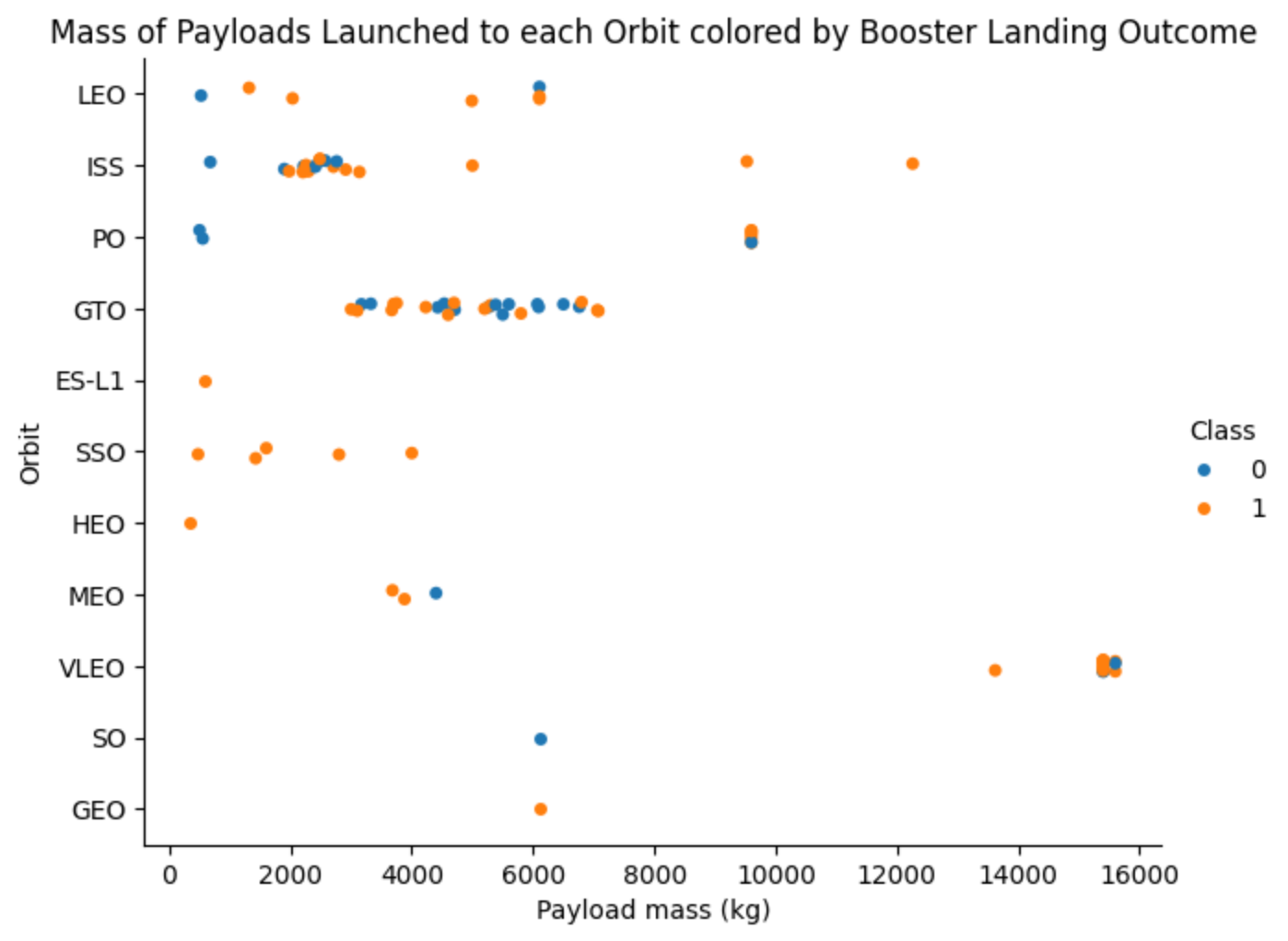
Flight Number vs. Orbit Type

- As with earlier, there were more success in later flights than in earlier ones.
- There also seemed to be a shift in target orbits among the later flights.



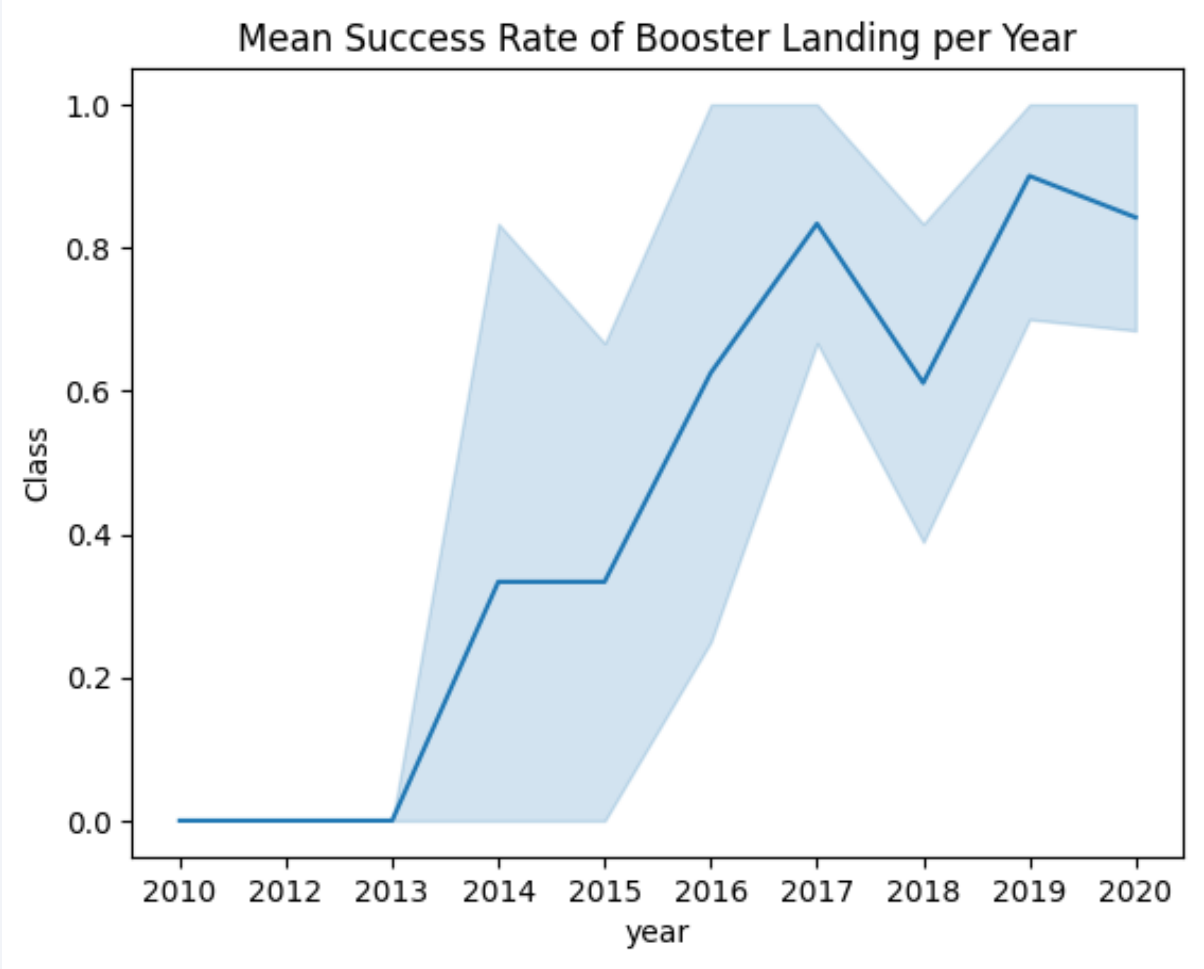
Payload vs. Orbit Type

- Booster landing success appeared to be concentrated in lighter payloads.
- But there were also fewer flights with more than 7000 kg payloads.
- Among those heavy-lift flights, the percentage of success appeared to be higher.



Launch Success Yearly Trend

- As time went on, landing the Falcon 9 boosters became more successful.
- Starting 2016, the success rate stayed above 50%.



All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Cape Canaveral Air Force Station (CCAFS) LC-40/SLC-40 – same site but SLC (Space Launch Complex) is the newer designation by the Air Force
- Vandenberg Air Force Base (VAFB)
- Kennedy Space Center (KSC)

Launch Site Names Begin with 'CCA'

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Among the flights launch from CCAFS seemed to be test or demo flights, like in row 1, where the payload was 'qualification unit' with zero mass, and the following rows were 'demo flight'.

Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

SUM(PAYLOAD_MASS__KG_)
45596

- Between 2010 and 2020, as provided by the dataset, NASA had already launched a total of 45,596 kg of payloads in their missions.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

AVG(PAYLOAD_MASS__KG_)
2928.4

- The average mass of payloads carried by the F9 v1.1 booster rocket was 2928.4 kg, with a range of 1316 to 4535 kg.

First Successful Ground Landing Date

```
SELECT MIN(Date) AS first_ground_landing FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

first_ground_landing
2015-12-22

- In this mission, the booster used was F9 FT B1019 carrying LEO satellites launched from CCAFS by Orbcomm. The rocket successfully landed on a ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ BETWEEN 4001 AND 5999;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- In the query, the upper and lower limits were adjusted considering the inclusivity of the BETWEEN operator.

Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) AS Count
FROM (SELECT
  CASE WHEN Mission_Outcome LIKE 'Success%' THEN 'Successful'
    ELSE 'Failure'
  END AS Mission_Outcome
  FROM SPACEXTABLE)
GROUP BY Mission_Outcome;
```

Mission_Outcome	Count
Failure	1
Successful	100

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	Count
Failure	1
Successful	100

- 'Mission' refers to the actual purpose of the flight. Not the landing of the booster.
- The sole failed mission was from NASA and seemed to had encountered an anomaly in flight. The booster was supposed to land on a drone ship.

Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ =
(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

- The heaviest payloads were at 15,600 kg and were all related to Starlink by SpaceX themselves.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
SELECT substr(Date, 6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE substr(Date,0,5)='2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- These missions in January and April of 2015 were resupply missions to the ISS conducted by SpaceX for NASA. The resupplies were successful but the booster failed to land on the drone ship.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- In this period, there was a total of 8 successful booster landings.

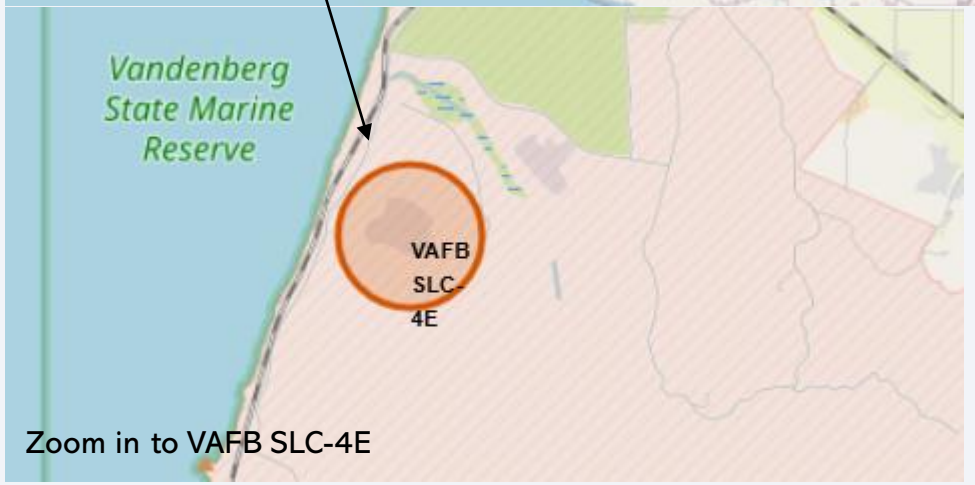
Section 3

Launch Sites Proximities Analysis

Launch Site Locations

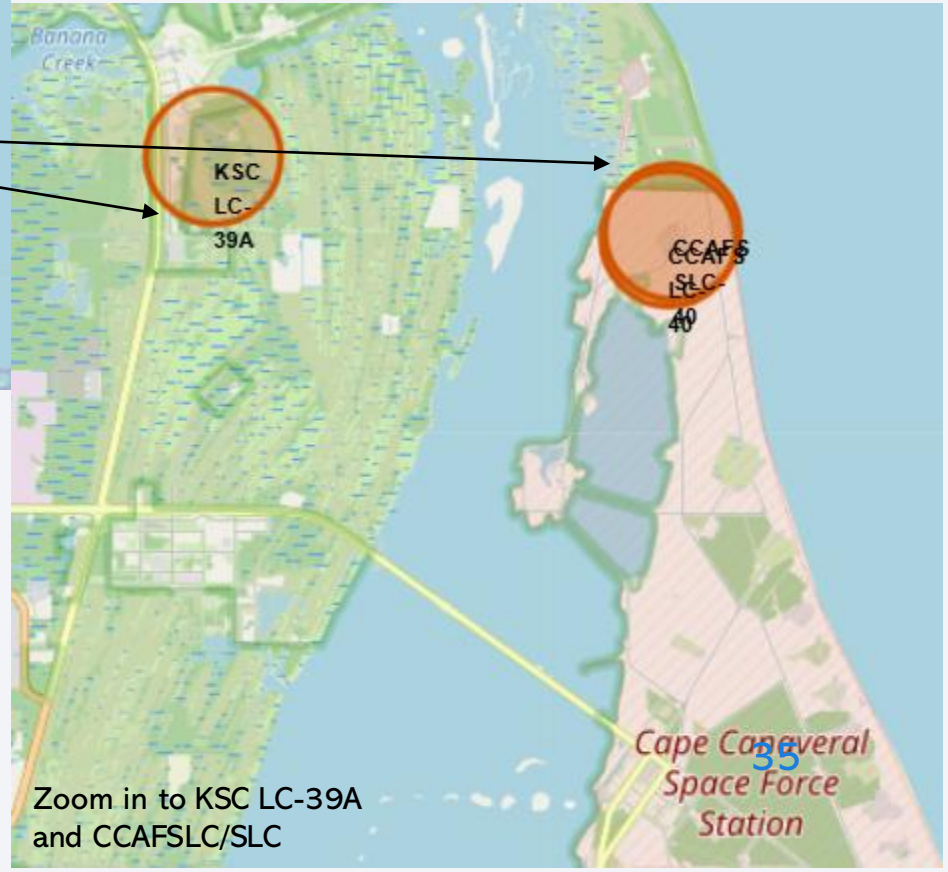
Global map

- With latitude coordinates between 28.5 and 34.6 for all sites, they are far from the equator.



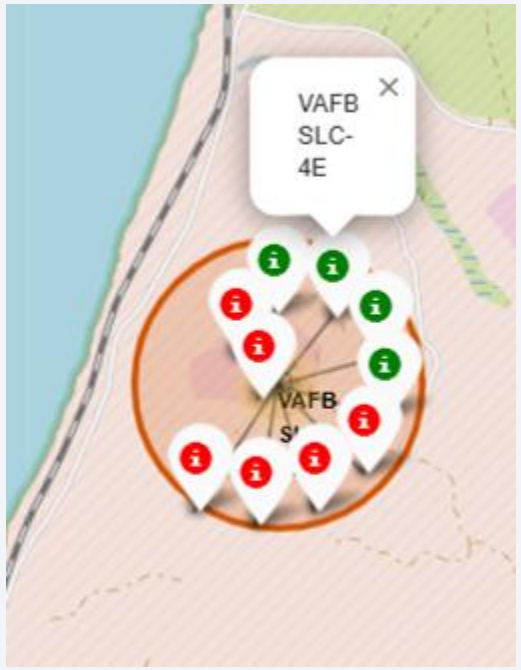
Zoom in to VAFB SLC-4E

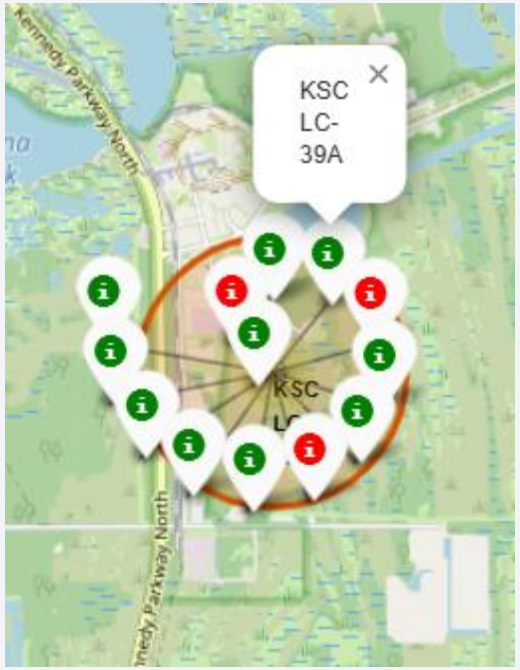
- The launch sites are very close to the coasts.

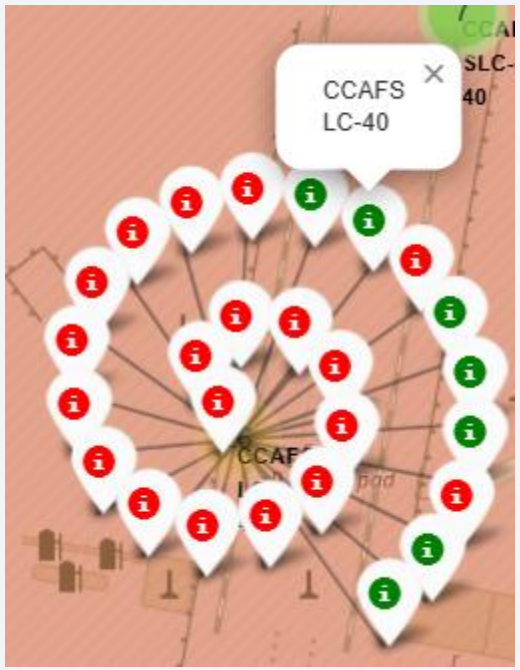


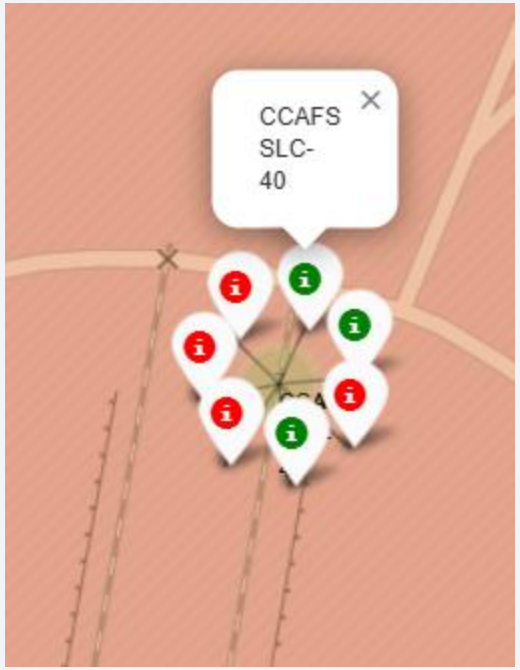
Zoom in to KSC LC-39A and CCAFSLC/SLC

Landing Outcomes from each Site



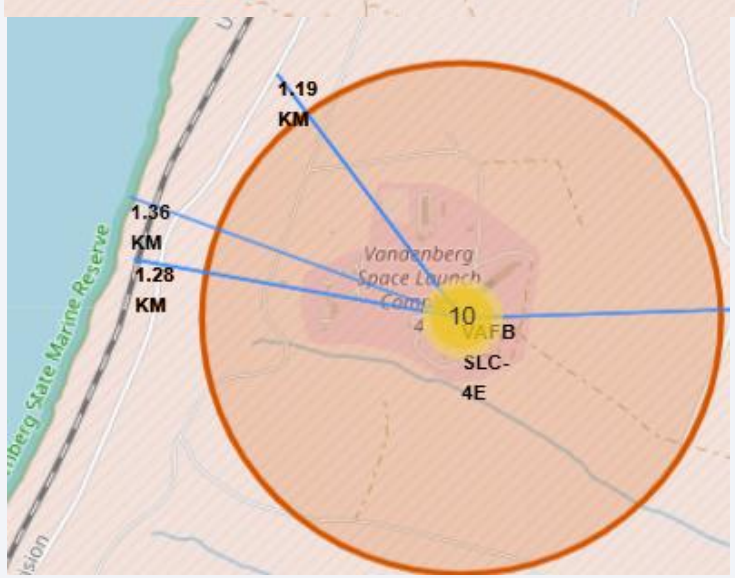






- CCAFS LC-40 had the most flights but had also the most failures in landing the Falcon 9 booster.
- KSC LC-39A had the most success in booster landing.

Launch Site and its Proximities

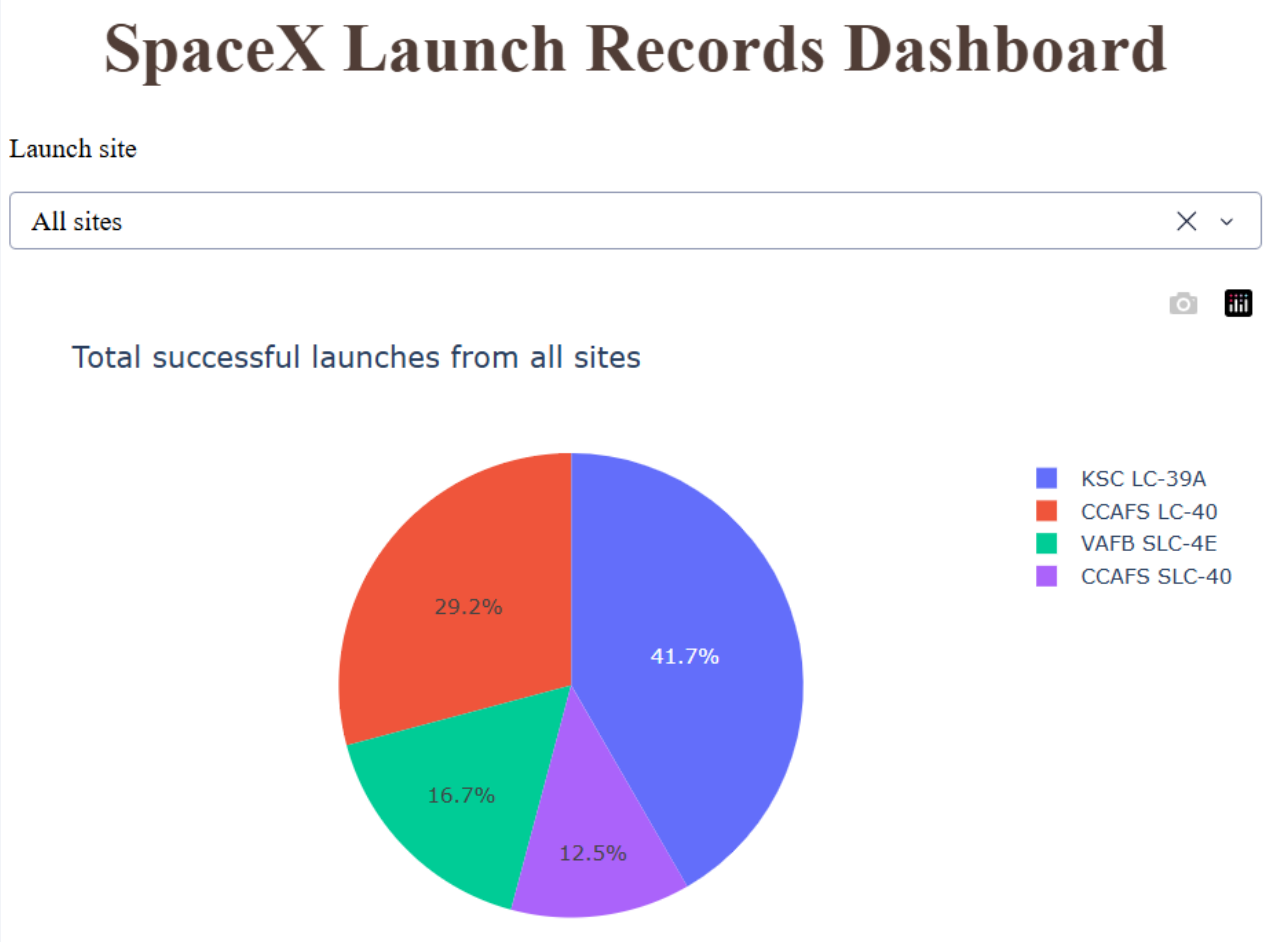


- Lompoc City, California is 14.09 km away from VAFB site. A railroad and a highway are 1.28 km and 1.19 km away, respectively. The coastline is 1.36 km away.
- Aside from being close to coastlines, launch sites are also near railroads and highways for accessibility, but are far away from cities or residential places to minimize risks or impact to the general public and the like.

Section 4

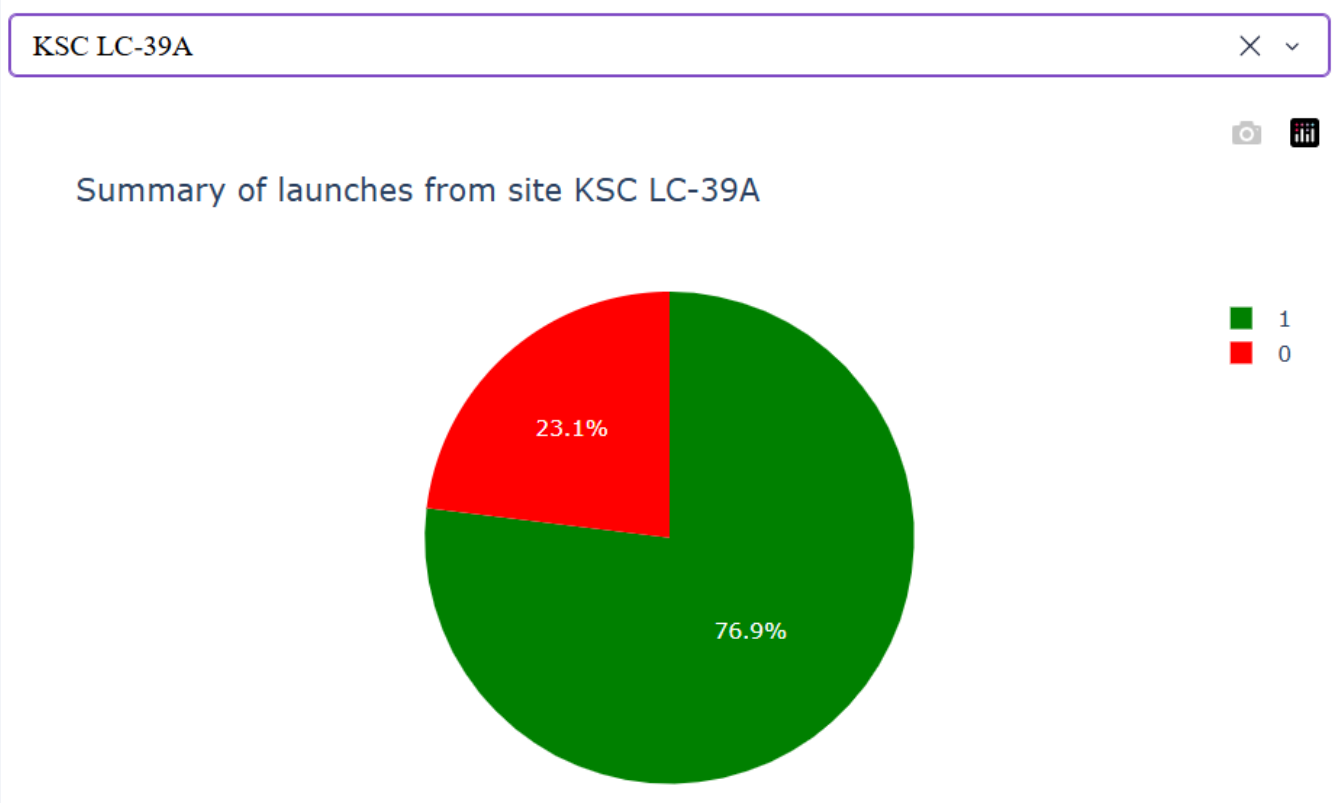
Build a Dashboard with Plotly Dash

Successful Launches from All Sites



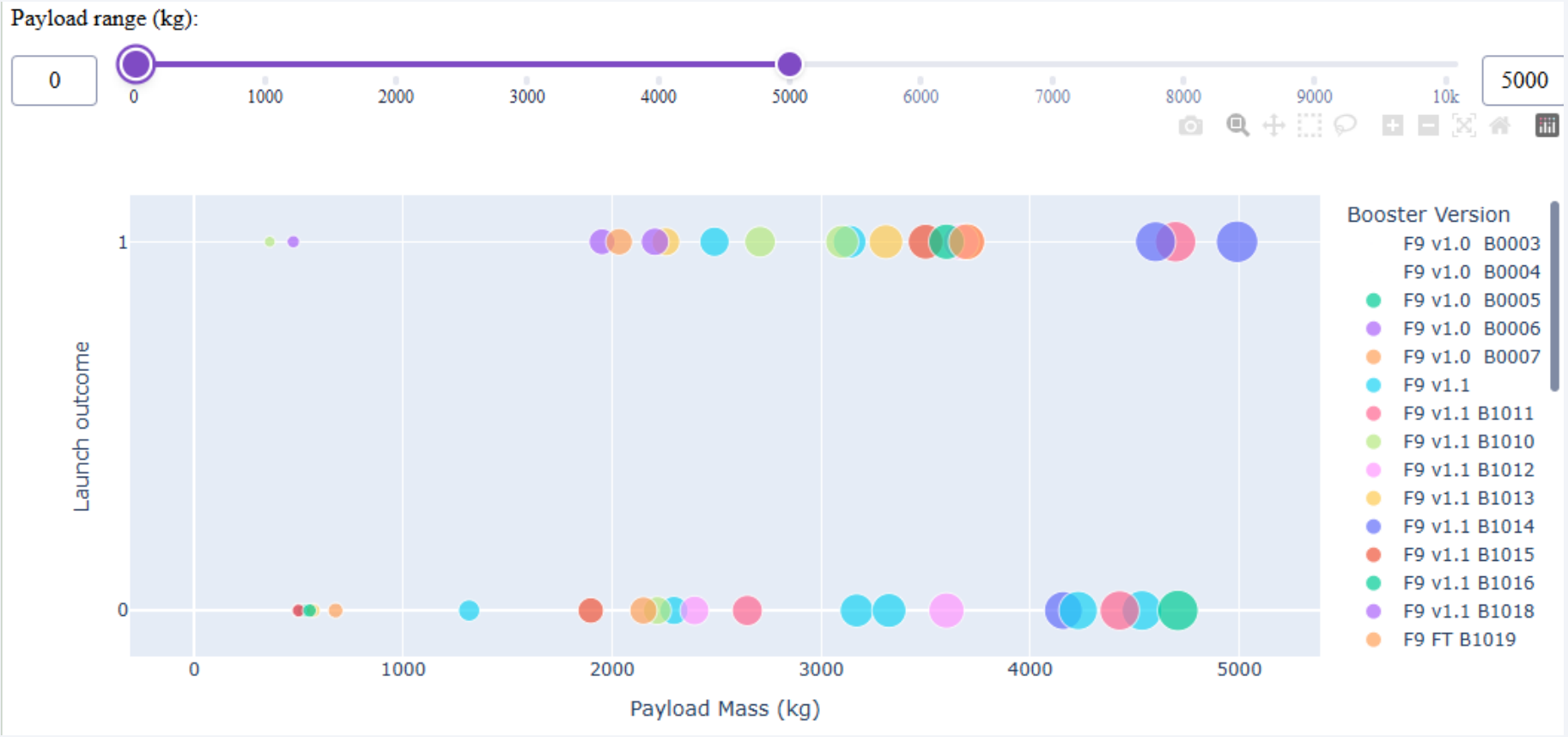
- KSC LC-39A had the highest contribution of successful launches.
- CCAFS SLC-40 had the lowest.
- However, when considering the CCAFS sites as one, their sum amounts to 41.7% of the total, similar with that of KSC LC-39A.

Launch Site with Highest Success Rate



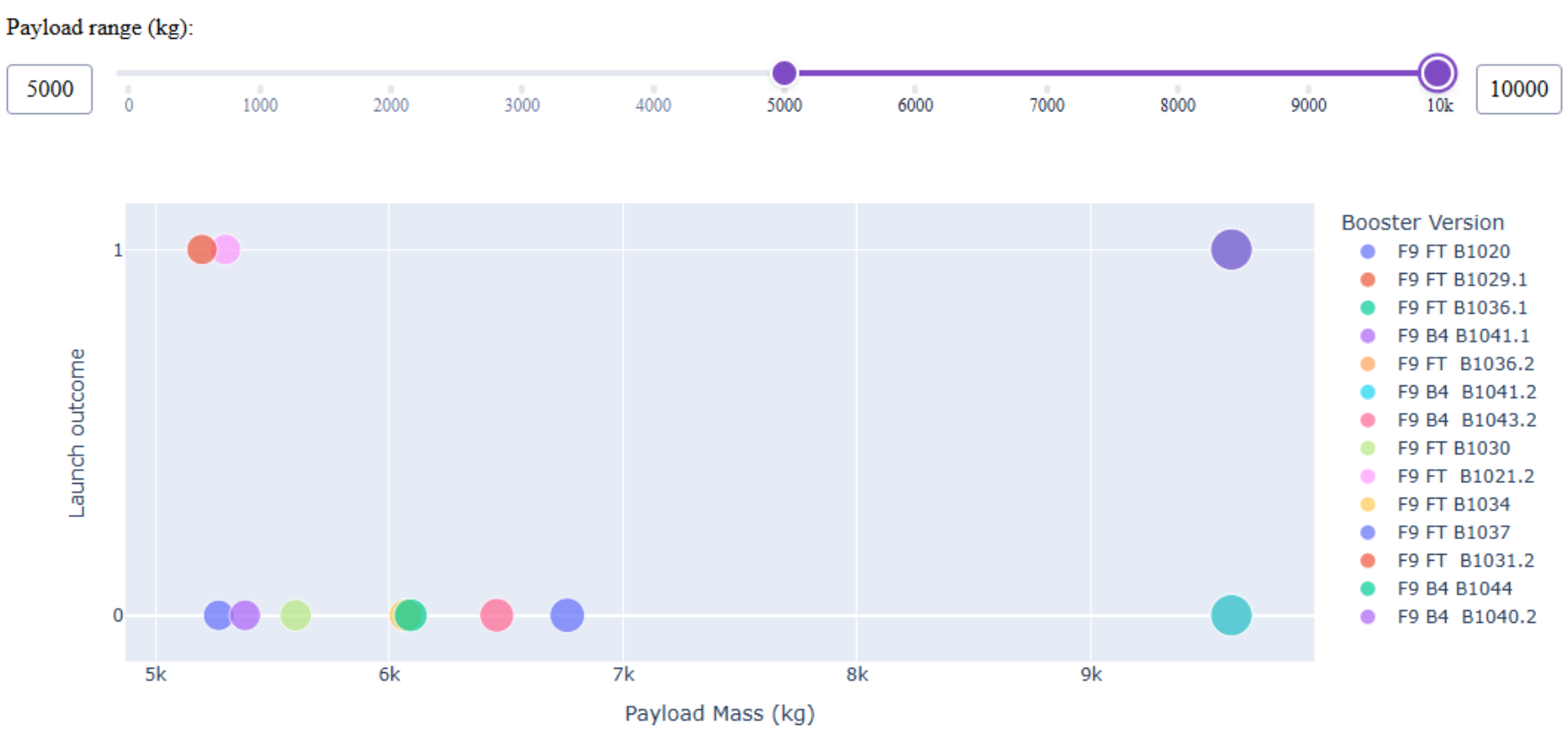
- Among the launches from KSC LC-39A, 76.9% were successful in landing the Falcon 9 booster.
- This translates to a 3.33 success-to-failure ratio

Payload Mass vs Outcome



- Success rate of lighter payloads is higher than the heavier ones.
- In the lower range, the predominant boosters are F9 v1.0 and v1.1

Payload Mass vs Outcome

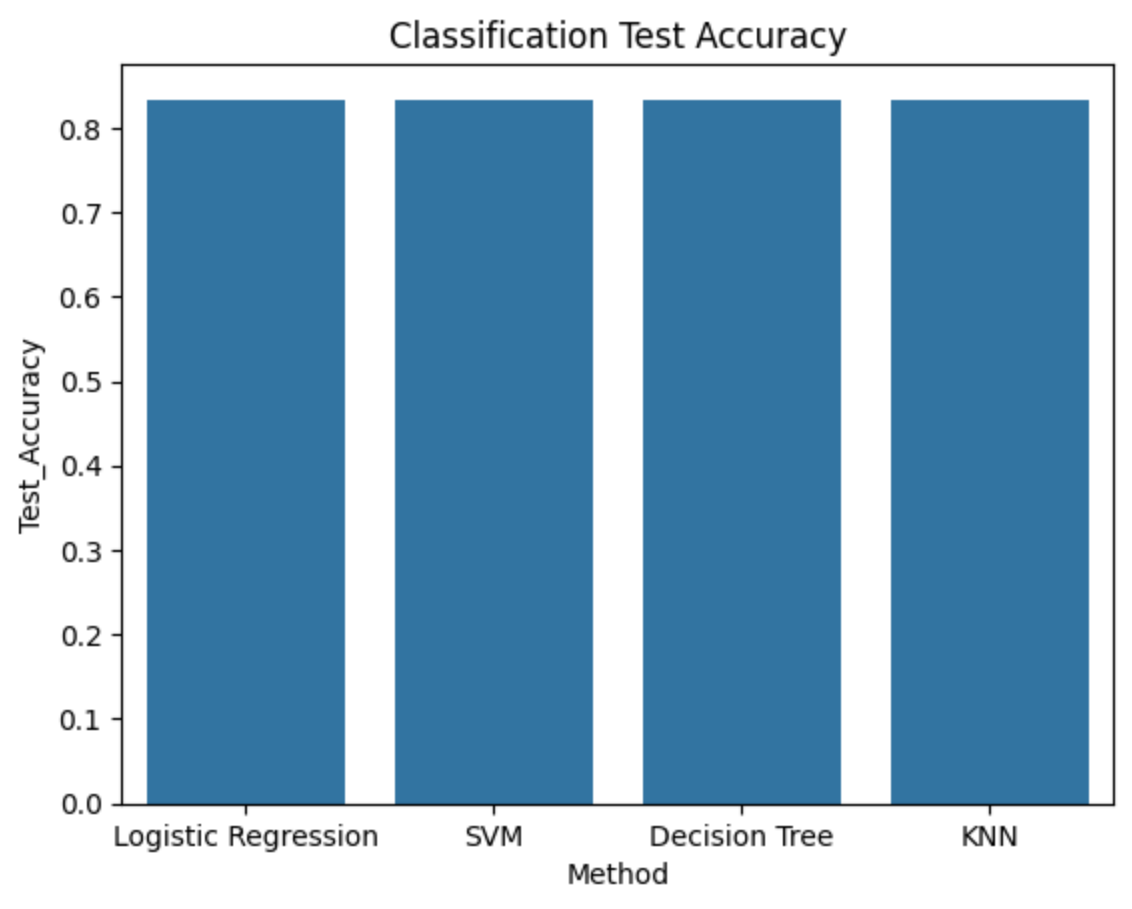


- In the higher range, the predominant boosters are F9 FT and B4

Section 5

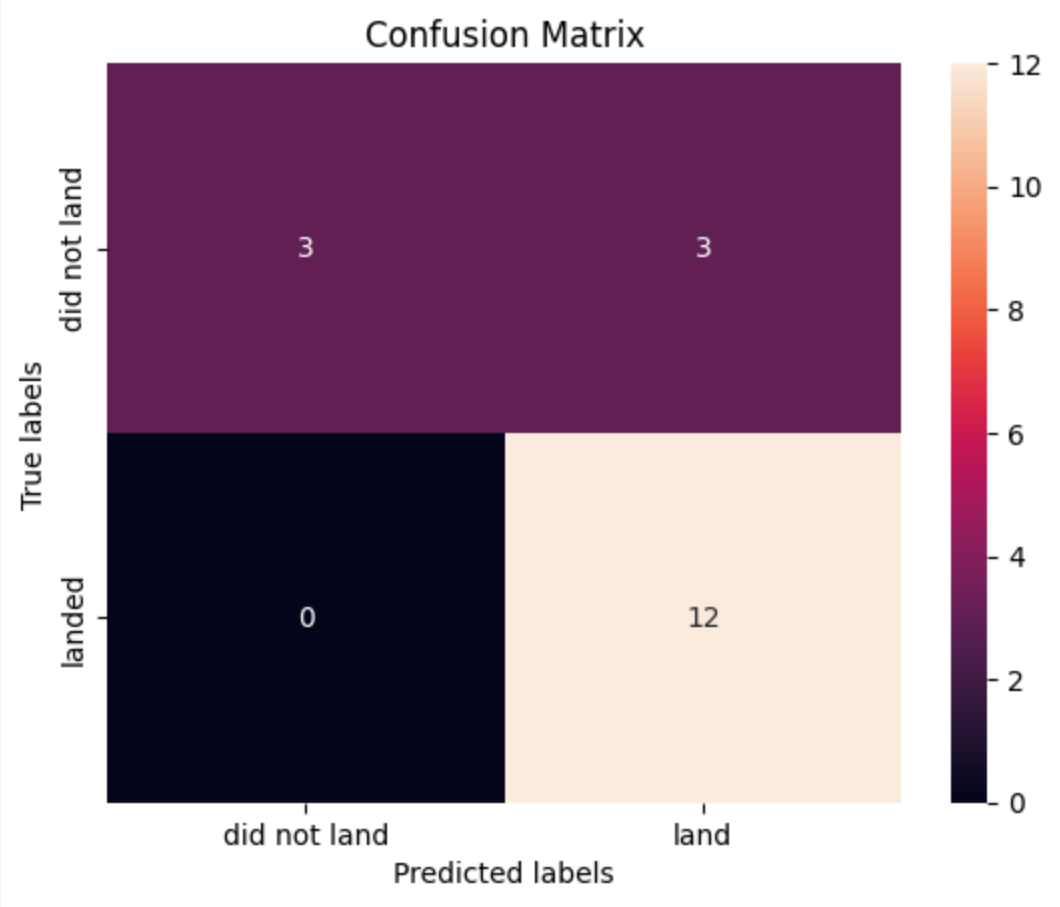
Predictive Analysis (Classification)

Classification Accuracy



The test accuracies were the same at 83.33%, so there is no best model to choose.

Confusion Matrix



The confusion matrices were also the same.

- 12 true positives
- 3 true negatives
- 3 false positives

Conclusions

- Landing the first stage of booster rockets became more successful as time went on, regardless of the payload mass, orbit type or launch site. Most probably due to ongoing research and technological advancements in the field.
- Launch sites are strategically located closer to coastlines, highways and railroads while being far away from cities or residential areas.
- All the classification algorithms used scored an accuracy of 83.33%. More data may be needed to develop predictive models, particularly testing data for accuracy evaluation.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!